

PVC injectors 20-40 kg/h

with diaphragm non-return valve

545-0615, 545-3615 / 545-0715, 545-3715;

545-06157, 545-36157 / 545-07157, 545-37157

Installation and operating instructions



English (GB) Installation and operating instructions

Original installation and operating instructions.

CONTENTS

	Page
1. Safety instructions	2
1.1 Symbols used in this document	2
1.2 Safety advice for handling chlorine	2
1.2.1 Hazards of chlorine gas	2
1.2.2 Hazards of liquid chlorine	2
1.2.3 Rules of conduct	2
1.3 Protective equipment	3
1.3.1 Personal protective equipment	3
1.3.2 Obligations of the operating authority	3
1.3.3 Storage of the protective equipment	3
2. General information	3
2.1 Correct usage	3
2.2 Inappropriate usage	3
2.3 Qualification of the users	3
2.4 Responsibilities of the operator	3
2.5 Maintenance and service personnel	3
2.6 Warranty	3
2.7 Transport and storage	3
2.8 Unpacking	3
3. Technical data	4
3.1 General technical data	4
3.2 Suction capacity / connection data	4
3.2.1 Injectors with a suction capacity up to 20 kg/h	4
3.2.2 Injectors with a suction capacity of 20 to 40 kg/h	4
3.3 Dimensioning of injectors	4
3.3.1 Suction capacity up to 20 kg/h Cl ₂	4
3.3.2 Suction capacity up to 10 kg/h NH ₃ , 20 kg/h SO ₂	4
3.3.3 Suction capacity 20 to 40 kg/h Cl ₂	4
3.3.4 Suction capacity 10-20 kg/h NH ₃ , 20-40 kg/h SO ₂	4
3.4 Dimensions	5
4. Structure and functioning	5
4.1 Functioning	5
4.2 Structure	5
5. Installation	6
5.1 Mounting the injector	6
5.2 Connecting the injector	6
6. Start-up	7
6.1 Checks before start-up	7
6.1.1 Checking the gas solution line for tightness	7
6.1.2 Checking the diaphragm non-return valve	7
6.1.3 Checking the functioning of the injector	7
7. Operation	7
8. Maintenance	8
8.1 Intervals for cleaning and maintenance	8
8.2 Maintenance kit	8
8.3 Assembly drawing	9
8.3.1 Injectors with a suction capacity up to 20 kg/h	9
8.3.2 Injectors with a suction capacity of 20 to 40 kg/h	10
8.4 Cleaning the nozzle	11
8.5 Checking the diaphragm non-return valve	11
8.5.1 Checking the closing mechanism	11
8.5.2 Replacing the piston	11
8.5.3 Replacing the seat and diaphragm	11
8.5.4 Checking the spring	11
8.5.5 Assembling the diaphragm non-return valve	11
9. Fault finding	12
10. Disposal	12

**Warning**

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Safety instructions

These installation and operating instructions contain important information for the user of the product:

- Technical data
- Instructions for commissioning and use
- Safety instructions

If you require further information, or if problems occur that are not described in detail in this manual, please contact Grundfos.

1.1 Symbols used in this document

This manual contains the following standardised safety instructions about possible residual risks:

**Warning**

If these safety instructions are not observed, it may result in personal injury.

Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment.

Note

Notes or instructions that make the job easier and ensure safe operation.

1.2 Safety advice for handling chlorine**1.2.1 Hazards of chlorine gas****Warning**

Toxic by inhalation.

Irritating to eyes, respiratory system and skin.

- Causes whooping cough, dyspnoea and lacrimation.
- Has a slight paralysing effect to the central nervous system.
- Concentrations of more than 10 ppm chlorine gas in the respiratory air mean an acute danger to life.
- Inhaling air with a high concentration of chlorine gas for a long time is lethal.

1.2.2 Hazards of liquid chlorine

- Causes burns of skin.
- Causes reddening and blistering of skin.

1.2.3 Rules of conduct

- Change chlorine containers only with gas mask.
- Enter contaminated rooms only with protective suit and compressed-air respirator.
- In case of flight wear gas mask, if possible. Observe wind direction.

Front page title:

PVC injectors 20-40 kg/h

Front page sub-title:

with diaphragm non-return valve

1.3 Protective equipment

1.3.1 Personal protective equipment

The operating authority of a chlorine gas dosing system has to provide respiratory equipment (full-sight gas mask), personally fitted, labelled by name, with an effective chlorine filter (B2P3) and at least 1 spare filter per gas mask, for each operator. The protective equipment must be stored easily accessible outside the chlorine rooms.

- For systems with chlorine drums at least 2 protective suits with compressed-air respirators must be available.

1.3.2 Obligations of the operating authority

- Introducing the operators to handling the protective equipment.
- The operators must use the personal protective equipment regularly, or carry out exercises with it at least every 6 months.
- The gas mask filters must be replaced regularly
 - after the expiry of the date of durability
 - at least 6 months after opening (note the opening date on the filter)
 - after contact with chlorine.
- Observe employing prohibition according to the local laws.

1.3.3 Storage of the protective equipment

- Outside the chlorine rooms.
- Well visible.
- Easily available at any time.
- Protected from dust and moisture.

2. General information

2.1 Correct usage

Injectors are used to inject Cl_2 , NH_3 and SO_2 gas into water in water treatment systems within the framework of the applications described in these operating instructions.

The media that may be used with each model are specified on the name plate.

2.2 Inappropriate usage

Operational safety is only guaranteed, if the product is used correctly. All operating methods conflicting with correct usage are not permitted, and lead to the expiry of all liability claims.

Warning



Unauthorised structural modifications to the product may result in serious damage to equipment and personal injury.

It is forbidden to open, modify, bridge, remove, bypass or disable components, especially safety equipment.

2.3 Qualification of the users

Based on his professional training and experience, the user must have a good knowledge of the operation of chlorine gas dosing systems. He must be so familiar with the relevant laws and regulations governing safety in the workplace and accident prevention, and with the recognised standards of good practice, that he is able to judge the safe working condition of a chlorine gas dosing system.

The user is responsible for operation and monitoring of the product at the installation site.

The responsibilities of the user:

- Read this manual before operating the product.
- Be trained by qualified Grundfos personnel in the operation of the product.
- Observe the relevant regulations governing safety in the workplace and accident prevention.
- Wear appropriate protective clothing in accordance with national regulations for the prevention of accidents when operating the system and handling chemicals.

2.4 Responsibilities of the operator

The operator of the product is responsible for the following:

- Ensure that this manual is kept clearly accessible in the immediate vicinity of the system for the entire service life.
- Meet the installation requirements specified by the manufacturer (environmental conditions, required water connections and fittings, electrical connection, protective pipe for dosing line if necessary, audible or optical warning device if necessary).
- Ensure that water lines and armatures are regularly checked, serviced and maintained.
- Obtain official approval for storing chemicals, if necessary.
- Train users in the operation of the system.
- Ensure that the regulations for the prevention of accidents are observed in the installation site.
- Provide all users and service personnel with protective clothing (e.g. face mask, gloves, protective apron) in accordance with the national regulations.

2.5 Maintenance and service personnel

The product may only be maintained and serviced by personnel authorised by Grundfos.

2.6 Warranty

The warranty period is 24 months from the date of delivery.

A guarantee claim in accordance with our general terms of sale and delivery is only valid if the following requirements are fulfilled:

- The product is used in accordance with the information within this manual.
- The product is not dismantled or handled incorrectly.
- The maintenance is carried out by authorised and qualified personnel.
- Only original spare parts are used for repairs during maintenance.
- Only parts permitted by Grundfos are used in the entire gas dosing system.

Typical wear parts are excluded from warranty, e. g.:

- gaskets, O-rings
- the injector nozzle, because the wear depends on the local water quality.

2.7 Transport and storage

- Transport carefully, do not throw!
- Store in a dry and cool place.

2.8 Unpacking

Observe when unpacking:

- The delivery must be complete.
- No humidity and no foreign matters should get into connections.
- Keep the injector in a horizontal position, so that the inserted nozzle cannot fall out.
- Mount the injector as soon as possible after unpacking.

3. Technical data

3.1 General technical data

Dosing medium	Cl ₂ , NH ₃ , SO ₂ and CO ₂
Motive water pressure	See section 3.3 Dimensioning of injectors
Counterpressure	
Permissible motive water temperature	+5 °C to +30 °C
Permissible ambient temperature	+5 °C to +40 °C
Gas dosing system	VGS 143

3.2 Suction capacity / connection data

3.2.1 Injectors with a suction capacity up to 20 kg/h

Injector	Maximum suction capacity			
	Cl ₂	NH ₃	SO ₂	CO ₂
545-0615 / 545-3615	20 kg/h	-	-	-
545-06157 / 545-36157	-	10 kg/h	20 kg/h	4.8 kg/h
Motive water connection (input, output)	DN 32			
Gas connection	DN 20			

3.2.2 Injectors with a suction capacity of 20 to 40 kg/h

Injector	Maximum suction capacity			
	Cl ₂	NH ₃	SO ₂	CO ₂
545-0715 / 545-3715	40 kg/h	-	-	-
545-07157 / 545-37157	-	20 kg/h	40 kg/h	9 kg/h
Motive water connection (input, output)	DN 50			
Gas connection	DN 20			

3.3 Dimensioning of injectors

The criteria for the selection of an injector are suction power, counterpressure and motive water pressure (admission pressure).

Usually the required suction power and counterpressure is generally specified. Other parameters such as motive water pressure and water quantity per hour can be found in the dimensioning table.

3.3.1 Suction capacity up to 20 kg/h Cl₂

Counterpressure [bar]	Motive water pressure [bar]	Water quantity [m ³ /h]	Injector
0.5	3	7	545-0615
1.2	4	8	
1.8	5	9	
2.3	6	9.6	
2.8	7	10.2	
3.3	8	11	
3.5	9	7.7	545-3615
4.2	10	8.3	
4.7	11	8.8	
5.3	12	9.3	
5.8	13	9.7	
6.3	14	10.1	
6.8	15	10.7	
7.3	16	11.3	

3.3.2 Suction capacity up to 10 kg/h NH₃, 20 kg/h SO₂

Counterpressure [bar]	Motive water pressure [bar]	Water quantity [m ³ /h]	Injector
0.5	3	7	545-06157
1.2	4	8	
1.8	5	9	
2.3	6	9.6	
2.8	7	10.2	
3.3	8	11	
3.5	9	7.7	
4.2	10	8.3	
4.7	11	8.8	545-36157
5.3	12	9.3	
5.8	13	9.7	
6.3	14	10.1	
6.8	15	10.7	
7.3	16	11.3	

3.3.3 Suction capacity 20 to 40 kg/h Cl₂

Counterpressure [bar]	Motive water pressure [bar]	Water quantity [m ³ /h]	Injector
0.5	3	13.2	545-0715
1.2	4	15.2	
1.8	5	16.8	
2.3	6	18.3	
2.8	7	19.7	
3.3	8	21	
3.5	9	15	
4.2	10	15.7	545-3715
4.7	11	16.5	
5.3	12	17.2	
5.8	13	18	
6.3	14	18.5	
6.8	15	19.2	
7.3	16	20	

3.3.4 Suction capacity 10-20 kg/h NH₃, 20-40 kg/h SO₂

Counterpressure [bar]	Motive water pressure [bar]	Water quantity [m ³ /h]	Injector
0.5	3	13.2	545-07157
1.2	4	15.2	
1.8	5	16.8	
2.3	6	18.3	
2.8	7	19.7	
3.3	8	21	
3.5	9	15	
4.2	10	15.7	545-37157
4.7	11	16.5	
5.3	12	17.2	
5.8	13	18	
6.3	14	18.5	
6.8	15	19.2	
7.3	16	20	

Note

All data refers to drinking water quality motive water.

Example

To achieve a suction power of 20 kg/h Cl_2 with a counterpressure of 2.8 bar, a motive water pressure (admission pressure) of 7 bar and a water quantity of 10.2 m³/h is required. See also table 3.3.1 Suction capacity up to 20 kg/h Cl_2 .

3.4 Dimensions

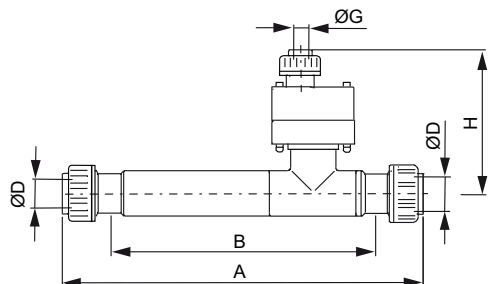


Fig. 1 Dimensioned drawing

Pos.	Description
A	Overall length
B	Fitting distance
H	Height

Injectors up to 20 kg/h

Injector	A [mm]	B [mm]	H [mm]
545-0615	480	350	190
545-3615			
545-06157			
545-36157			

Injectors up to 40 kg/h

Injector	A [mm]	B [mm]	H [mm]
545-0715	710	540	320
545-3715			
545-07157			
545-37157			

4. Structure and functioning

4.1 Functioning

By reducing the line cross-section with a nozzle, the motive water is accelerated and creates a negative pressure. The gas is drawn in by the negative pressure.

The gas is then whirled by the diffuser and mixed with the motive water in the downstream mixing section to produce operating water.

4.2 Structure

Components of the injector:

- Gas supply connection (1)
- Body with diaphragm non-return valve (2): The diaphragm non-return valve prevents water from entering the gas-conveying parts of the device.
- Motive water inlet connection (3)
- The nozzle (4) accelerates the motive water flow and creates a negative pressure.
- The diffuser with mixing section (5) mixes motive water with gas.
- Gas solution line connection = motive water outlet (6)

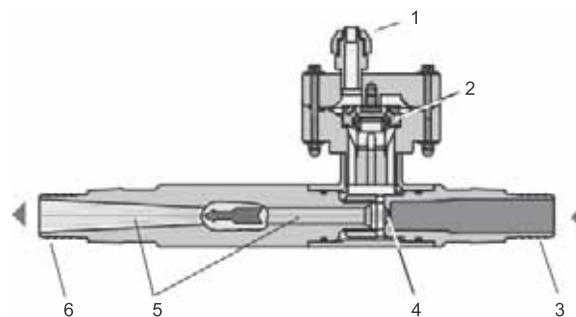


Fig. 2 Functioning scheme

5. Installation

The following conditions must be fulfilled to ensure proper functioning:

- Ensure sufficient motive water and motive water pressure (admission pressure). Fit a booster pump if necessary.
- Keep the solution line between the injector and the injection unit as short as possible.
- Use clean water, install a strainer if necessary.
- Choose the length and cross-section of the gas supply line according to the values proposed in the table below. For the calculation of the values, a pressure loss of 50 mbar was assumed, without considering a possible effect on the safety of the system.

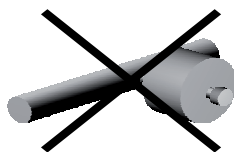
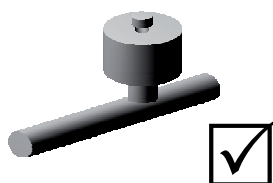
Gas supply line [m]	Dosing quantity [g/h]									
	40	100	250	500	1000	2000	4000	10000	20000	40000
0	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 15	DN 15	DN 20
10	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 15	DN 15	DN 20
20	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15	DN 20	DN 25
30	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15	DN 20	DN 25
40	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15	DN 20	DN 25
50	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15	DN 15	DN 20	DN 25
75	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15	DN 20	DN 25	DN 32
100	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15	DN 20	DN 25	DN 32

5.1 Mounting the injector



Warning

When mounting, the diaphragm non-return valve must be vertical on injector.



TM03 6626 2911

Fig. 3 Position of diaphragm non-return valve

Caution Observe flow direction (arrow sticker).

Caution Mount and connect the injector free of stresses.

Fit the injector to the wall or other suitable location using the fixing material.

The fixing material is supplied with injectors of a suction capacity up to 20 kg/h. It includes two clamps, two M6 x 60 screws and two S10 dowels (product number 95701436).

For injectors with a suction capacity of 20 to 40 kg/h the fixing material is optional:

Description	Content	Product No.
Optional fixing material for injectors with a suction capacity of 20 to 40 kg/h	2 consoles	96729319
	4 hexagon nuts	
	4 washers	
	2 pipe clamps	
	2 threaded pins	
	4 dowels	
	4 wood screws	

5.2 Connecting the injector

1. Connect motive water supply line (pos. 3, fig. 2).
2. Connect gas solution line (6).
3. Connect gas supply line (1).

6. Start-up

Caution

For commissioning, all components of the entire system must be ready for operation.

Observe the manuals of the other components.

6.1 Checks before start-up

Check the tightness of the entire system before start-up.



Warning

Only check for tightness when the entire system is ready for operation.

Observe the manuals of the other components.

Risk of gas leak.

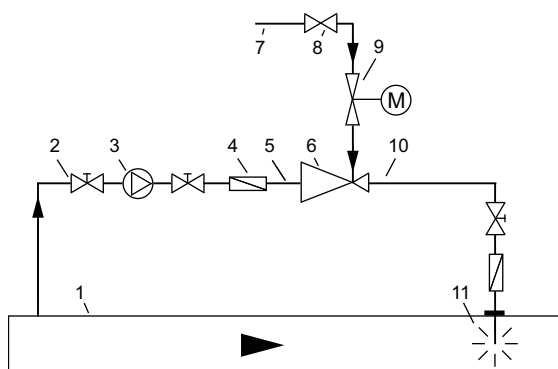
6.1.1 Checking the gas solution line for tightness



Warning

The system must be switched off. All container valves must be closed.

1. Loosen the union nut (pos. 100, fig. 5, 6) on the gas connection.
2. Disconnect the gas supply.
3. Open the motive water valve.
4. Open the injection unit isolating valve.
5. Switch on the motive water pump. If water escapes from between the motive water pump (pos. 3, fig. 4) and the injection unit (9), there is a leak in the line.
6. Switch off the motive water pump.
7. Close the motive water valve.
8. Close the injection unit isolating valve.
9. Repair the leak.
10. Repeat check for tightness.
11. If no water escapes, the gas supply can be reconnected.



TM03 6622 2911

Fig. 4 System and components

Pos.	Description
1	Supply line
2	Isolating valve
3	Motive water pump
4	Non-return valve
5	Motive water inlet injector
6	Injector
7	Gas supply line
8	Safety pressure relief valve
9	Gas dosing system
10	Gas solution line (motive water outlet injector)
11	Injection unit

6.1.2 Checking the diaphragm non-return valve

1. Loosen the union nut (pos. 100, fig. 5, 6) on the gas connection.
2. Disconnect the gas supply.
3. Open the motive water valve.
4. Open the injection unit isolating valve.
5. Switch on the motive water pump
6. Briefly shut the injection unit isolating valve. If water escapes from the gas connection, the diaphragm non-return valve is defective.
7. Switch off the motive water pump.
8. Close the motive water valve.
9. Repair the diaphragm non-return valve.
10. Repeat the check. If no water escapes, the diaphragm non-return valve is working properly.
11. Reconnect the gas supply.

6.1.3 Checking the functioning of the injector

1. Loosen the union nut (pos. 100, fig. 5, 6) on the gas connection.
2. Disconnect the gas supply.
3. Open the motive water valve.
4. Open the injection unit isolating valve.
5. Switch on the motive water pump.
6. Place your thumb over the gas connection. If suction can clearly be felt, the injector is working correctly. If not, continue with the next point.
7. Check the admission pressure (water quantity).
8. Check the counterpressure.
9. Check the line length and cross-sections, see section 5. *Installation*.
10. If the fault cannot be eliminated, contact the next Grundfos service centre.

7. Operation

Note

The injector is passive. Adjustment has to be made at other components of the system.

Read the manuals of the other components before start-up.

8. Maintenance



Warning
*Before starting cleaning and maintenance work,
 switch off the entire system.
 Danger of gas escape!*



Warning
*Before re-starting, check the tightness.
 Danger of gas escape!*

8.1 Intervals for cleaning and maintenance

Component	Normal use ¹⁾	Heavy duty ²⁾
Upstream dirt trap	every 12 weeks	every 8 weeks
Nozzles	every 12 weeks	every 8 weeks
Diaphragm non-return valve, piston, seat and diaphragm	every 6 months	every 6 months

¹⁾ Admission pressure up to 6 bar, high motive water quality (drinking water).

²⁾ Admission pressure more than 6 bar, poor motive water quality.

Note

In case of poor water quality, extreme pressure conditions, or faults, the maintenance intervals are shorter.

8.2 Maintenance kit

We recommend to keep maintenance kits in stock to ensure that worn parts can be replaced quickly when necessary.

The parts included in a maintenance kit are marked with an asterisk in the legends of the assembly drawings, see section 8.3 *Assembly drawing*.

Injectors	Maintenance kit, depending on the medium	Product No.
Suction capacity up to 20 kg/h	for Cl ₂	96688740
	for NH ₃ , SO ₂ and CO ₂	96688739
Suction capacity 20 to 40 kg/h	for Cl ₂	96688741
	for NH ₃ , SO ₂ and CO ₂	96688742

8.3 Assembly drawing

8.3.1 Injectors with a suction capacity up to 20 kg/h

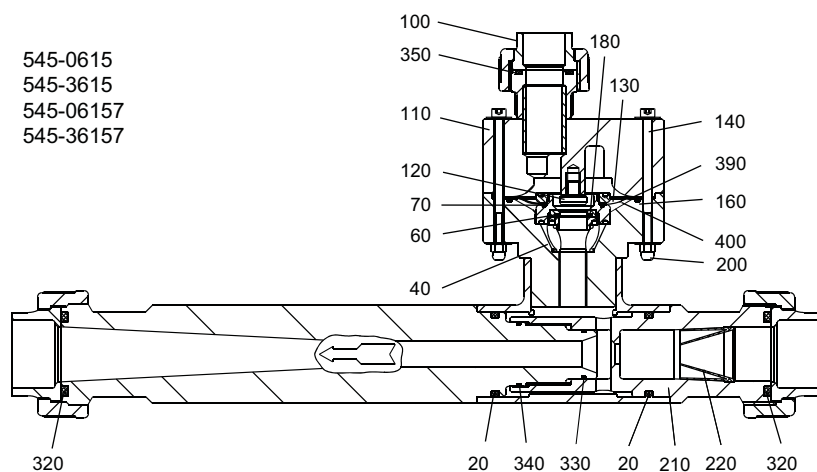


Fig. 5 Assembly drawing

Pos.	Description
020	Two O-rings (38 x 46)*
040	Spring
060	Seat*
070	O-ring (32 x 36)*
100	Union nut gas supply line
110	Top part
120	Piston*
130	Diaphragm*
140	Cylinder screws
160	O-ring (72 x 77)*
180	O-ring (17 x 22)*
210	Nozzle
220	Swirl part
320	Two O-rings (40 x 51)*
330	O-ring (24 x 28)*
340	O-ring (32 x 36)*
350	O-ring (28 x 35)*
390	Diaphragm*
400	Diaphragm*

* Included in maintenance kit

TM03 6629 2911

8.3.2 Injectors with a suction capacity of 20 to 40 kg/h

545-0715
545-3715
545-07157
545-37157

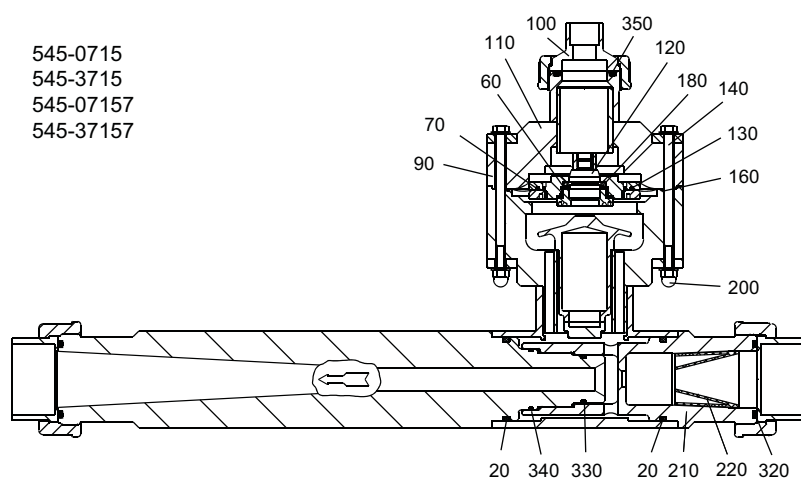


Fig. 6 Assembly drawing

Pos.	Description
20	Two O-rings (60 x 70)*
60	Seat*
70	O-ring (80 x 86)*
100	Union nut gas supply line
110	Top part
120	Piston*
130	Separating diaphragm*
140	Cylinder screws
160	O-ring (135 x 140)*
180	O-ring (28 x 35)*
200	Cap nut
210	Nozzle
220	Swirl part
320	Two O-rings (60 x 70)*
330	O-ring (33 x 40)*
340	O-ring (45 x 51)*
350	O-ring (47 x 57)*

* Included in maintenance kit

8.4 Cleaning the nozzle

Deposits in the nozzle prevent the injector from creating the required negative pressure, leading to a drop in suction capacity. We recommend to use hydrochloric acid for cleaning the nozzle.



Warning

Observe the regulations for the use of hydrochloric acid.

Caution Do not use sharp objects for cleaning.

1. Disconnect the gas supply.
2. Loosen the water line (on injectors with a tube connection, loosen the tube) and remove the injector.
3. Unscrew the nozzle (pos. 210, fig. 5, 6). Leave the swirl part (220) in the nozzle.
4. Clean the nozzle with hydrochloric acid.
5. Reassemble and fit the injector in the reverse order.

8.5 Checking the diaphragm non-return valve

8.5.1 Checking the closing mechanism

1. Close the adjustment valve on the dosing system.
2. Loosen the union nut (pos. 100, fig. 5, 6).
3. Disconnect the gas supply.
4. Briefly close the isolating valve behind the injector several times. If no water escapes, the diaphragm non-return valve is working properly. If water escapes, the piston and seat must be replaced.
5. Replace piston (120) and seat (60).
6. Open the isolating valve.
7. Connect the gas supply line.
8. Set the adjustment valve on the dosing system to the required dosing flow.

8.5.2 Replacing the piston

Do not use tools when screwing in the piston.

Caution If the piston surface is damaged, the closing mechanism may not work.

1. Shut down the system.
2. Close the isolating valve behind the injector.
3. Loosen the cap nut (pos. 200, fig. 5, 6).
4. Remove the cylinder screws (140).
5. Remove the top part (90).
6. Unscrew the piston (120).
7. Screw in a new piston (by hand), replace O-ring (160) if necessary.
8. Slide the diaphragm disc (with diaphragm, seat, screw part and O-rings) onto the piston.
 - Turn the diaphragm disc so that the holes line up with the drill holes in the upper part
 - Do not lose the spring (40).
9. Fit the top part and diaphragm disc.
10. Slightly tighten the screws in a criss-cross sequence.

8.5.3 Replacing the seat and diaphragm

1. Shut down the system.
2. Close the isolating valve behind the injector.
3. Loosen the cap nut (pos. 200, fig. 5, 6).
4. Remove the cylinder screws (140).
5. Remove the top part (110).

Replacing the diaphragm

1. Unscrew the diaphragm ring (pos. 80, fig. 7 or 8).
2. Check the O-ring (pos. 70), replace O-Ring if necessary.
3. Injectors up to 20 kg/h: remove diaphragm (130, 400, 390); injectors up to 40 kg/h: remove diaphragm (130).
4. Insert a new diaphragm.

Caution Observe the mounting position.

5. Firmly tighten the diaphragm ring (80).

Replacing the seat

1. Unscrew the screw part (50).
2. Remove the seat (60) with O-ring.
3. Insert a new seat with O-ring.

Caution Observe the mounting position.

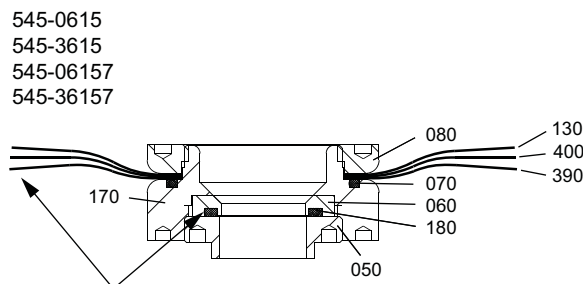


Fig. 7 Position of diaphragm, injectors up to 20 kg/h

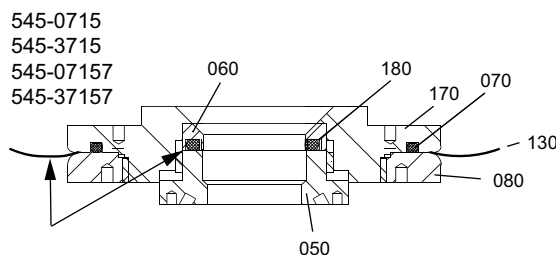


Fig. 8 Position of diaphragm, injectors up to 40 kg/h

4. Tighten the diaphragm ring so that the seat can be pushed firmly yet smoothly over the piston.

8.5.4 Checking the spring

The spring is plastic-coated for protection against corrosive gases.

Caution Replace the spring, if the coating is damaged.

8.5.5 Assembling the diaphragm non-return valve

1. Slide the complete diaphragm disc (with diaphragm, seat, screw part and O-rings) onto the piston.

Caution Observe the mounting position.

2. Turn the diaphragm disc so that the holes line up with the drill holes in the upper part.
3. Insert the spring.
4. Fit the top part and diaphragm disc.
5. Fit the cylinder screws and slightly tighten them in a criss-cross sequence.
6. Check the closing mechanism.
7. If water escapes, re-tighten the diaphragm ring slightly.
8. If water still escapes, re-tighten the diaphragm ring again, until it closes tightly.

TM03 6627 2911

TM03 6628 2911

9. Fault finding

Fault	Cause	Remedy
1. Maximum dosing flow is not achieved despite correctly selected injector.	a) Swirl part in the nozzle damaged or not present.	Insert a new swirl part.
	b) Counterpressure on the injector is too high (solution outlet too long or line cross-section too small).	Install the injector closer to the injection unit, observe the line cross-sections.
	c) Motive water pressure or quantity not correctly set.	Check the settings using the table in section 3.3 <i>Dimensioning of injectors</i> .
	d) Dynamic pressure on the injector too low due to line loss.	Check length and cross-section of motive water and solution line, use the least possible number of elbows.
	e) Process water temperature higher than 30 °C.	Adjust process water temperature to approx. 20-25 °C.
	f) The dynamic pressure created in the injector is too low due to faulty gas supply.	Replace the gas supply line.

10. Disposal

This product and all its associated parts must be disposed of in an environmentally friendly manner. Use appropriate waste collection services. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

Argentina

Bombas GRUNDFOS de Argentina S.A.
Ruta Panamericana km. 37.500 Lote 34A
1619 - Garin
Pcia. de Buenos Aires
Phone: +54-3327 414 444
Telefax: +54-3327 411 111

Australia

GRUNDFOS Pumps Pty. Ltd.
P.O. Box 2040
Regency Park
South Australia 5942
Phone: +61-8-8461-4611
Telefax: +61-8-8340 0155

Austria

GRUNDFOS Pumpen Vertrieb Ges.m.b.H.
Grundfosstraße 2
A-5082 Grödig/Salzburg
Tel.: +43-6246-883-0
Telefax: +43-6246-883-30

Belgium

N.V. GRUNDFOS Bellux S.A.
Boomsesteenweg 81-83
B-2630 Aartselaar
Tel.: +32-3-870 7300
Télécopie: +32-3-870 7301

Belorussia

Представительство ГРУНДФОС в
Минске
220123, Минск,
ул. В. Хоружей, 22, оф. 1105
Тел.: +(37517) 233 97 65
Факс: (37517) 233 9769
E-mail: grundfos_minsk@mail.ru

Bosnia/Herzegovina

GRUNDFOS Sarajevo
Trg Heroja 16,
BiH-71000 Sarajevo
Phone: +387 33 713 290
Telefax: +387 33 659 079
e-mail: grundfos@bih.net.ba

Brazil

BOMBAS GRUNDFOS DO BRASIL
Av. Humberto de Alencar Castelo Branco,
630
CEP 09850 - 300
São Bernardo do Campo - SP
Phone: +55-11 4393 5533
Telefax: +55-11 4343 5015

Bulgaria

Grundfos Bulgaria EOOD
Slatina District
Iztocna Tangenta street no. 100
BG - 1592 Sofia
Tel. +359 2 49 22 200
Fax. +359 2 49 22 201
email: bulgaria@grundfos.bg

Canada

GRUNDFOS Canada Inc.
2941 Brighton Road
Oakville, Ontario
L6H 6C9
Phone: +1-905 829 9533
Telefax: +1-905 829 9512

China

Grundfos Alldos
Dosing & Disinfection
ALLDOS (Shanghai) Water Technology
Co. Ltd.
West Unit, 1 Floor, No. 2 Building (T 4-2)
278 Jinhu Road, Jin Qiao Export Process-
ing Zone
Pudong New Area
Shanghai, 201206
Phone: +86 21 5055 1012
Telefax: +86 21 5032 0596
E-mail: grundfosalldos-CN@grundfos.com

China

GRUNDFOS Pumps (Shanghai) Co. Ltd.
22 Floor, Xin Hua Lian Building
755-775 Huai Hai Rd, (M)
Shanghai 200020
PRC
Phone: +86-512-67 61 11 80
Telefax: +86-512-67 61 81 67

Croatia

GRUNDFOS CROATIA d.o.o.
Cebini 37, Buzin
HR-10010 Zagreb
Phone: +385 1 6595 400
Telefax: +385 1 6595 499
www.grundfos.hr

Czech Republic

GRUNDFOS s.r.o.
Čapkovského 21
779 00 Olomouc
Phone: +420-585-716 111
Telefax: +420-585-716 299

Denmark

GRUNDFOS DK A/S
Martin Bachs Vej 3
DK-8850 Bjerringbro
Tlf.: +45-87 50 50 50
Telefax: +45-87 50 51 51
E-mail: info_GDK@grundfos.com
www.grundfos.com/DK

Estonia

GRUNDFOS Pumps Eesti OÜ
Peterburi tee 92G
11415 Tallinn
Tel: + 372 606 1690
Fax: + 372 606 1691

Finland

OY GRUNDFOS Pumput AB
Mestarintie 11
FIN-01730 Vantaa
Phone: +358-3066 5650
Telefax: +358-3066 56550

France

Pompes GRUNDFOS Distribution S.A.
Parc d'Activités de Chesnes
57, rue de Malacombe
F-38290 St. Quentin Fallavier (Lyon)
Tél.: +33-4 74 82 15 15
Télécopie: +33-4 74 94 10 51

Germany

GRUNDFOS Water Treatment GmbH
Reetzstraße 85
D-76327 Pfingsttal (Söllingen)
Tel.: +49 7240 61-0
Telefax: +49 7240 61-177
E-mail: gwt@grundfos.com

Germany

GRUNDFOS GMBH
Schlüterstr. 33
D-40699 Erkrath
Tel.: +49-(0) 211 929 69-0
Telefax: +49-(0) 211 929 69-3799
E-mail: infoservice@grundfos.de
Service in Deutschland:
E-mail: kundendienst@grundfos.de

Greece

GRUNDFOS Hellas A.E.B.E.
20th km. Athinon-Markopoulou Av.
P.O. Box 71
GR-19002 Peania
Phone: +0030-210-66 83 400
Telefax: +0030-210-66 46 273

Hong Kong

GRUNDFOS Pumps (Hong Kong) Ltd.
Unit 1, Ground floor
Siu Wai Industrial Centre
29-33 Wing Hong Street &
68 King Lam Street, Cheung Sha Wan
Kowloon
Phone: +852-27861706 / 27861741
Telefax: +852-27858664

Hungary

GRUNDFOS Hungária Kft.
Park u. 8
H-2045 Törökbálint,
Phone: +36-23 511 110
Telefax: +36-23 511 111

India

GRUNDFOS Pumps India Private Limited
118 Old Mahabalipuram Road
Thoraiakkam
Chennai 600 096
Phone: +91-44 2496 6800

Indonesia

PT GRUNDFOS Pompa
Jl. Rawa Sumur III, Blok III / CC-1
Kawasan Industri, Pulogadung
Jakarta 13930
Phone: +62-21-460 6909
Telefax: +62-21-460 6910 / 460 6901

Ireland

GRUNDFOS (Ireland) Ltd.
Unit A, Merrywell Business Park
Ballymount Road Lower
Dublin 12
Phone: +353-1-4089 800
Telefax: +353-1-4089 830

Italy

GRUNDFOS Pompe Italia S.r.l.
Via Gran Sasso 4
I-20060 Truccazzano (Milano)
Tel.: +39-02-95838112
Telefax: +39-02-95309290 / 95838461

Japan

GRUNDFOS Pumps K.K.
Gotanda Metalion Bldg. 5F,
5-21-15, Higashi-gotanda
Shiagawa-ku, Tokyo,
141-0022 Japan
Phone: +81 35 448 1391
Telefax: +81 35 448 9619

Korea

GRUNDFOS Pumps Korea Ltd.
6th Floor, Aju Building 679-5
Yeoksam-dong, Kangnam-ku, 135-916
Seoul, Korea
Phone: +82-2-5317 600
Telefax: +82-2-5633 725

Latvia

SIA GRUNDFOS Pumps Latvia
Deglava biznesa centrs
Augusta Deglava ielā 60, LV-1035, Rīga,
Tālr.: + 371 714 9640, 7 149 641
Fakss: + 371 914 9646

Lithuania

GRUNDFOS Pumps UAB
Smolensko g. 6
LT-03201 Vilnius
Tel: + 370 52 395 430
Fax: + 370 52 395 431

Malaysia

GRUNDFOS Pumps Sdn. Bhd.
7 Jalan Peguam U1/25
Glenmarie Industrial Park
40150 Shah Alam
Selangor
Phone: +60-3-5569 2922
Telefax: +60-3-5569 2866

México

Bombas GRUNDFOS de México S.A. de
C.V.
Boulevard TLC No. 15
Parque Industrial Stiva Aeropuerto
Apodaca, N.L. 66600
Phone: +52-81-8144 4000
Telefax: +52-81-8144 4010

Netherlands

GRUNDFOS Netherlands
Veluwezoom 35
1326 AE Almere
Postbus 22015
1302 CA ALMERE
Tel.: +31-88-478 6336
Telefax: +31-88-478 6332
E-mail: info_gnl@grundfos.com

New Zealand

GRUNDFOS Pumps NZ Ltd.
17 Beatrice Tinsley Crescent
North Harbour Industrial Estate
Albany, Auckland
Phone: +64-9-415 3240
Telefax: +64-9-415 3250

Norway

GRUNDFOS Pumper A/S
Strømsveien 344
Postboks 235, Leirdal
N-1011 Oslo
Tlf.: +47-22 90 47 00
Telefax: +47-22 32 21 50

Poland

GRUNDFOS Pompy Sp. z o.o.
ul. Klonowa 23
Baranowo k. Poznania
PL-62-081 Przeźmierowo
Tel: (+48-61) 650 13 00
Fax: (+48-61) 650 13 50

Portugal

Bombas GRUNDFOS Portugal, S.A.
Rua Calvet de Magalhães, 241
Apartado 1079
P-2770-153 Paço de Arcos
Tel.: +351-21-440 76 00
Telefax: +351-21-440 76 90

România

GRUNDFOS Pompe România SRL
Bd. Biruintei, nr 103
Pantelimon county Ilfov
Phone: +40 21 200 4100
Telefax: +40 21 200 4101
E-mail: romanian@grundfos.ro

Russia

ООО Грундфос
Россия, 109544 Москва, ул. Школьная
39
Тел. (+7) 495 737 30 00, 564 88 00
Факс (+7) 495 737 75 36, 564 88 11
E-mail grundfos.moscow@grundfos.com

Serbia

GRUNDFOS Predstavništvo Beograd
Dr. Milutina Ivkovića 2a/29
YU-11000 Beograd
Phone: +381 11 26 47 877 / 11 26 47 496
Telefax: +381 11 26 48 340

Singapore

GRUNDFOS (Singapore) Pte. Ltd.
24 Tuas West Road
Jurong Town
Singapore 638381
Phone: +65-6865 1222
Telefax: +65-6861 8402

Slovenia

GRUNDFOS d.o.o.
Šlandrova 8b, SI-1231 Ljubljana-Črnuče
Phone: +386 1 568 0610
Telefax: +386 1 568 0619
E-mail: slovenia@grundfos.si

South Africa

Grundfos (PTY) Ltd.
Corner Mountjoy and George Allen Roads
Wilbart Ext. 2
Bedfordview 2008
Phone: (+27) 11 579 4800
Fax: (+27) 11 455 6066
E-mail: lsmart@grundfos.com

Spain

Bombas GRUNDFOS España S.A.
Camino de la Fuentequilla, s/n
E-28110 Algete (Madrid)
Tel.: +34-91-848 8800
Telefax: +34-91-628 0465

Sweden

GRUNDFOS AB
(Box 333) Lunnagårdsgatan 6
431 24 Mölndal
Tel.: +46(0)771-32 23 00
Telefax: +46(0)31-331 94 60

Switzerland

GRUNDFOS ALLDOS International AG
Schönmatzstraße 4
CH-4153 Reinach
Tel.: +41-61-717 5555
Telefax: +41-61-717 5500
E-mail: grundfosalldos-CH@grundfos.com

Switzerland

GRUNDFOS Pumpen AG
Bruggacherstrasse 10
CH-8117 Fällanden/ZH
Tel.: +41-1-806 8111
Telefax: +41-1-806 8115

Taiwan

GRUNDFOS Pumps (Taiwan) Ltd.
7 Floor, 219 Min-Chuan Road
Taichung, Taiwan, R.O.C.
Phone: +886-4-2305 0868
Telefax: +886-4-2305 0878

Thailand

GRUNDFOS (Thailand) Ltd.
92 Chaloe Phrakiat Rama 9 Road,
Dokmai, Pravej, Bangkok 10250
Phone: +66-2-725 8999
Telefax: +66-2-725 8998

Turkey

GRUNDFOS POMPA San. ve Tic. Ltd. Sti.
Gebze Organize Sanayi Bölgesi
İhsan dede Caddesi,
2. yol 200. Sokak No. 204
41490 Gebze/ Kocaeli
Phone: +90 - 262-679 7979
Telefax: +90 - 262-679 7905
E-mail: satis@grundfos.com

Ukraine

ТОВ ГРУНДФОС УКРАЇНА
01010 Київ, Вул. Московська 86,
Тел.: (+38 044) 390 40 50
Факс.: (+38 044) 390 40 59
E-mail: ukraine@grundfos.com

United Arab Emirates

GRUNDFOS Gulf Distribution
P.O. Box 16768
Jebel Ali Free Zone
Dubai
Phone: +971-4- 8815 166
Telefax: +971-4-8815 136

United Kingdom

GRUNDFOS Pumps Ltd.
Grovebury Road
Leighton Buzzard/Beds. LU7 8TL
Phone: +44-1525-850000
Telefax: +44-1525-850011

U.S.A.

GRUNDFOS Pumps Corporation
17100 West 118th Terrace
Olathe, Kansas 66061
Phone: +1-913-227-3400
Telefax: +1-913-227-3500

Uzbekistan

Представительство ГРУНДФОС в
Ташкенте
700000 Ташкент ул.Усмана Носира 1-й
тулик 5
Телефон: (3712) 55-68-15
Факс: (3712) 53-36-35

96681311 0911
Repl. 96681311 0907

GB

ECM: 1079563

The name Grundfos, the Grundfos logo, and the payoff Be-Think-Innovate are registered trademarks owned by Grundfos Management A/S or Grundfos A/S, Denmark. All rights reserved worldwide.
